

Breast Cancer Diagnosis Using Mammogram Images with CLAHE and Deep Learning Algorithms

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Simple Summary: Breast cancer is a serious health issue that affects millions of women worldwide. Early and accurate diagnosis is very important to improve survival rates. In this study, we used two types of computer algorithms—K-Nearest Neighbors (K-NN) and Convolutional Neural Networks (CNN)—to help detect breast cancer from ultrasound images. K-NN is a simple method that works well for small datasets, while CNN is a powerful deep learning technique that can automatically learn features from images. We tested both methods and found that CNN models performed slightly better than K-NN. The best model reached 97.84% accuracy, which shows great potential for helping doctors detect breast cancer early and more accurately, especially in areas with limited medical resources.

Abstract: Breast cancer remains a leading cause of mortality among women globally, emphasizing the need for efficient and accurate diagnostic solutions. This study investigates the classification of breast cancer using both K-Nearest Neighbors (K-NN) and Convolutional Neural Networks (CNN), leveraging the Breast Ultrasound Images Dataset. The objective is to compare the interpretability of traditional machine learning and the feature extraction power of deep learning for early detection. The dataset was preprocessed through grayscale conversion, resizing to 128×128 pixels, and feature extraction using Histogram of Oriented Gradients (HOG) for K-NN. CNN models including VGG16 and VGG19 were fine-tuned using pre-trained weights. Results show that the optimized K-NN model achieved an accuracy of 96%, while VGG16 and VGG19 reached 97.51% and 96.69% accuracy, respectively. Additionally, integrating CNN with CLAHE preprocessing boosted performance to 97.84%. The study confirms that CNNs outperform K-NN in classification accuracy due to their hierarchical feature learning, although K-NN remains useful for smaller datasets with limited computational resources. These findings demonstrate the complementary strengths of both approaches and highlight the potential of AI-driven tools in delivering scalable, non-invasive diagnostic support for breast cancer detection in clinical and resource-constrained settings.

Keywords: KNN; CNN; ResNet50; CNN+CLAHE; VGG16; VGG19; breast cancer

1. Introduction

Breast cancer is one of the most prevalent and life-threatening diseases affecting women worldwide, posing a significant global health challenge. According to global cancer statistics, it accounts for approximately 24.5% of all cancer cases among women, making it the most commonly diagnosed malignancy in this demographic. The burden of breast cancer is not confined to any single region; its impact is evident in both developed and developing countries. However, disparities in healthcare access and diagnostic tools often result in delayed diagnoses, particularly in underprivileged regions, contributing to higher mortality rates in such areas.

The early detection of breast cancer plays a pivotal role in improving patient outcomes. When identified at an early stage, breast cancer is often more manageable and treatable, significantly increasing survival rates and reducing long-term healthcare costs. In this regard, the use of mammography has become the gold standard for breast cancer screening, offering detailed images that can aid in identifying abnormalities in breast tissue. However, interpreting mammograms remains a complex and time-consuming task that requires a high level of expertise, often unavailable in resource-limited settings.

To address these challenges, advancements in artificial intelligence (AI) and machine learning (ML) offer promising solutions for automating and enhancing the diagnostic process. This project explores the development of a breast cancer classification model using mammogram images, employing two complementary machine learning approaches: K-Nearest Neighbors (KNN) and Convolutional Neural Networks (CNN). KNN, a traditional algorithm, provides a straightforward yet powerful method for classifying data based on its proximity to labeled examples, while CNNs, a subset of deep learning, excel at extracting intricate features from image data, making them ideal for medical imaging tasks.

By integrating these approaches, the study seeks to achieve a balance between simplicity and advanced computational capabilities. The primary objective is to enhance diagnostic accuracy and efficiency while ensuring the solution remains accessible and scalable. Moreover, the project emphasizes the potential for such technologies to bridge healthcare gaps in underprivileged regions, enabling timely and reliable breast cancer detection even in resource-constrained environments.

By leveraging AI-powered tools, this research aims to contribute to global efforts in reducing breast cancer-related mortality and improving the quality of life for affected individuals (Zhu et al., 2022; Das et al., 2023).

A visual representation of the relationship between machine learning and deep learning, highlighting the key algorithms used in this study, is shown in Figure 1.

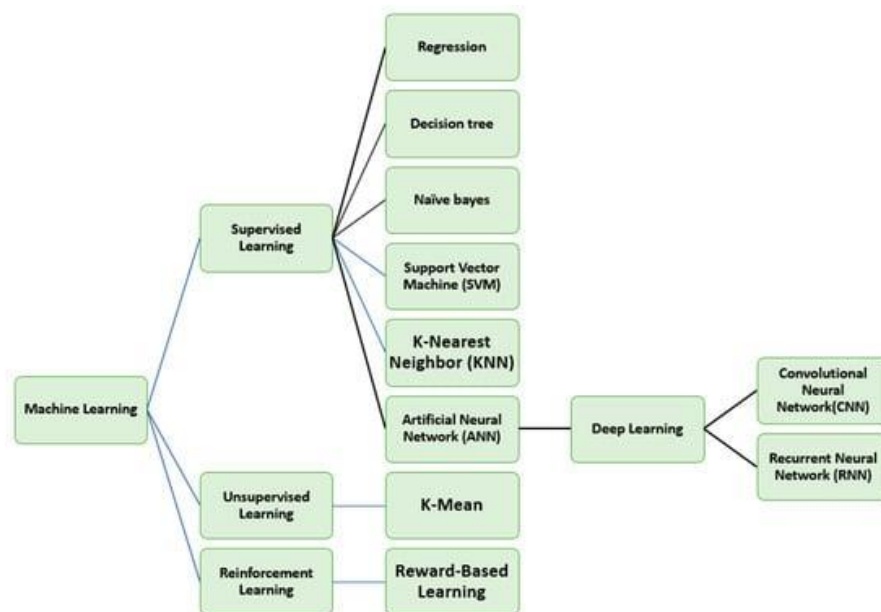


Figure 1. The Hierarchy of Machine Learning and Deep Learning.

2. Materials and Methods

2.1. Dataset and Data Collection

This study utilized 10,430 mammogram images obtained from the Mendeley and DDSM public repositories. The images were labeled by radiologists as either “Cancer” or “Non-Cancer” and served as the ground truth for supervised learning. Each image was converted to grayscale for simplicity and resized to 128×128 pixels to maintain consistency in model input dimensions. Histogram equalization was applied to improve contrast, followed by normalization to scale pixel values between 0 and 1. A train-test split of 80:20 was used to train and evaluate all models (Figure 2).

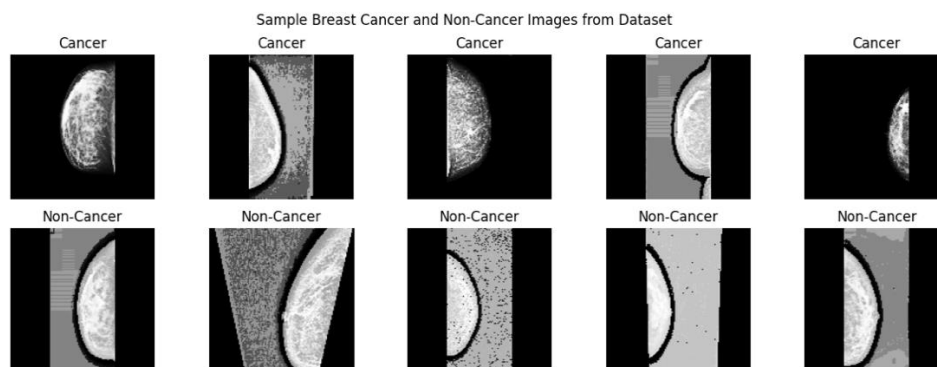


Figure 2. Grayscale Conversion of Mammogram Images.

2.2. Preprocessing

Preprocessing steps included:

- Grayscale conversion to reduce complexity.
- Resizing to 128×128 pixels.
- Histogram equalization to enhance contrast.
- Normalization of pixel values to $[0,1]$.
- Train-test split: 80% training and 20% testing.

2.3. Model Architecture and Configuration

We compared two approaches:

A. K-Nearest Neighbors (KNN):

The KNN model relied on Histogram of Oriented Gradients (HOG) for feature extraction. Euclidean distance was used to measure similarity, and a grid search was performed to find the optimal number of neighbors. The best accuracy was obtained with $K = 3$. The full training pipeline is shown in Figure 3, with accuracy trends by K-value in Figure 4. The confusion matrix illustrating model performance is shown in Figure 5.

```
# Train and evaluate KNN model
print("Training KNN model...")
knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train, y_train)
y_pred = knn_model.predict(X_test)

Training KNN model...
```

Figure 3. KNN Training Process.

```

# Accuracy vs. K value plot
acc = []
print("Generating accuracy vs. K-value plot...")
for i in range(1, 40):
    neigh = KNeighborsClassifier(n_neighbors=i).fit(X_train, y_train)
    yhat = neigh.predict(X_test)
    accuracy = metrics.accuracy_score(y_test, yhat)
    acc.append(accuracy)

plt.figure(figsize=(10, 6))
plt.plot(range(1, 40), acc, color='blue', linestyle='dashed', marker='o', markerfacecolor='red', markersize=10)
plt.title('Accuracy vs. K Value')
plt.xlabel('K-value')
plt.ylabel('Accuracy')
plt.show()

# Find maximum accuracy and the corresponding K value
max_accuracy = max(acc)
rounded_accuracy = round(max_accuracy, 2) # Adjust decimal precision if desired
optimal_k = acc.index(max_accuracy) + 1

print("Maximum accuracy (rounded):", rounded_accuracy, "at K =", optimal_k)

```

Figure 4. Accuracy vs. K-Value.

```

# Confusion matrix
print("Displaying confusion matrix...")
conf_matrix = metrics.confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues', xticklabels=['Non-Cancer', 'Cancer'], yticklabels=['Non-Cancer', 'Cancer'])
plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.show()

```

Figure 5. KNN Confusion Matrix.

B. Convolutional Neural Networks (CNN):

The CNN architecture leveraged three variants: VGG16, VGG19, and ResNet50, all initialized with pre-trained ImageNet weights. These models were fine-tuned for 25 epochs using the Adam optimizer and categorical cross-entropy loss. Data augmentation (rotation, zoom, flips) was applied to prevent overfitting. The workflow of the entire system is presented in Figure 6, while the high-level comparison between KNN and CNN processes is illustrated in Figure 7.

An additional variant was tested using CLAHE (Contrast Limited Adaptive Histogram Equalization) to enhance image clarity. This improved the visual contrast and, in turn, the CNN model's performance by helping it detect critical edges and textures more effectively.

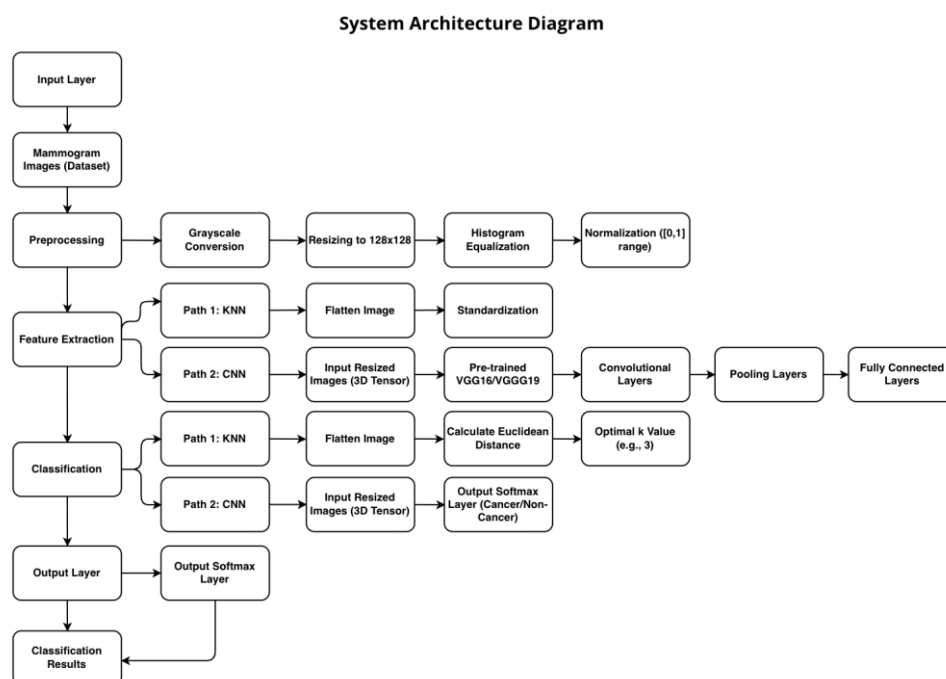


Figure 6. System Architecture Diagram for Breast Cancer Classification Workflow.

AI Model Diagram

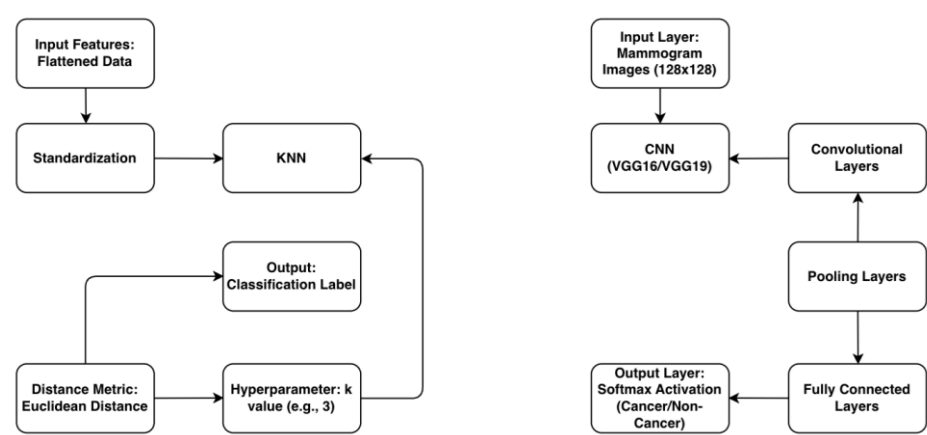


Figure 7. AI Model Diagram showing KNN and CNN branches

3. Results

3.1. KNN Model Performance

The KNN model, using HOG features and $K = 3$, achieved an overall accuracy of 96% (Figure 8). The performance metrics showed strong classification between benign and malignant classes, with relatively low false positives and negatives. The confusion matrix (Figure 10) further confirmed KNN's solid performance, although its capability is limited by the lack of automated feature learning. Performance trends across different K -values are shown in Figure 9.

```
# Evaluate the model
accuracy = metrics.accuracy_score(y_test, y_pred)
print(f"Accuracy of the KNN model after optimization: {accuracy * 100:.2f}%")

Accuracy of the KNN model after optimization: 95.30%
```

Figure 8. Accuracy of the KNN Model.

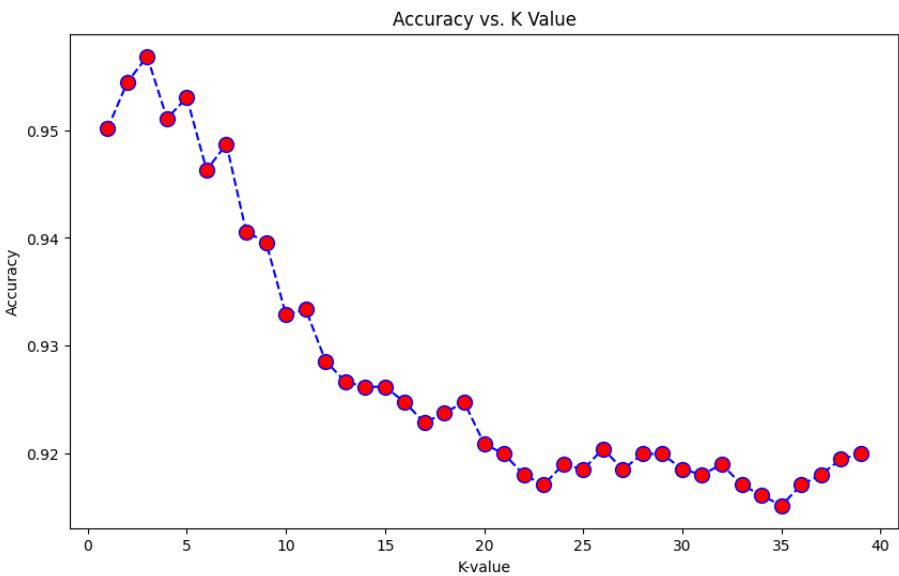


Figure 9. Accuracy vs. K Value.

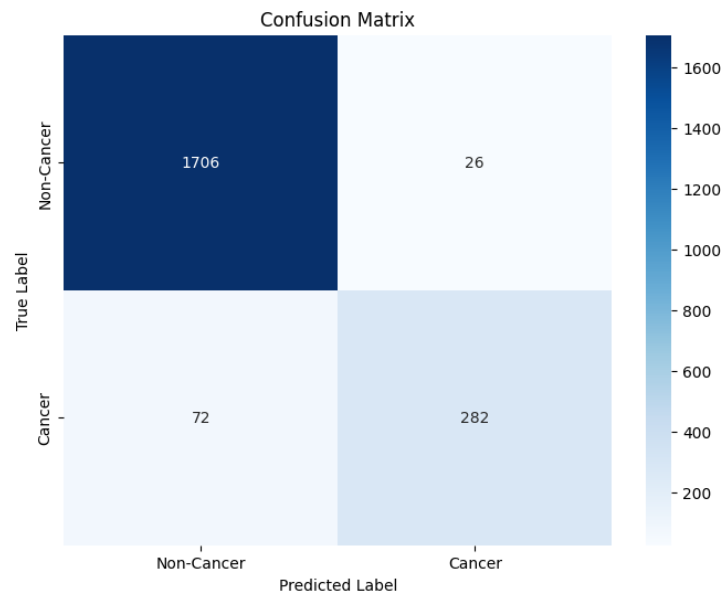


Figure 10. Confusion Matrix of KNN.

3.2. CNN Model Performance

- VGG16 achieved the best result, reaching 97.51% accuracy. Training and validation metrics are visualized in Figure 11, and the confusion matrix is shown in Figure 12. Its loss and accuracy curves, depicted in Figure 13, indicate good convergence with minimal overfitting.
- VGG19 followed with 96.69% accuracy, as shown in Figure 14, supported by its confusion matrix in Figure 15 and loss/accuracy curves in Figure 16.
- CNN with CLAHE preprocessing further improved performance, reaching 97.84% accuracy (Figure 17).
- ResNet50 produced a competitive result of 95.01% accuracy, as shown in Figure 18.

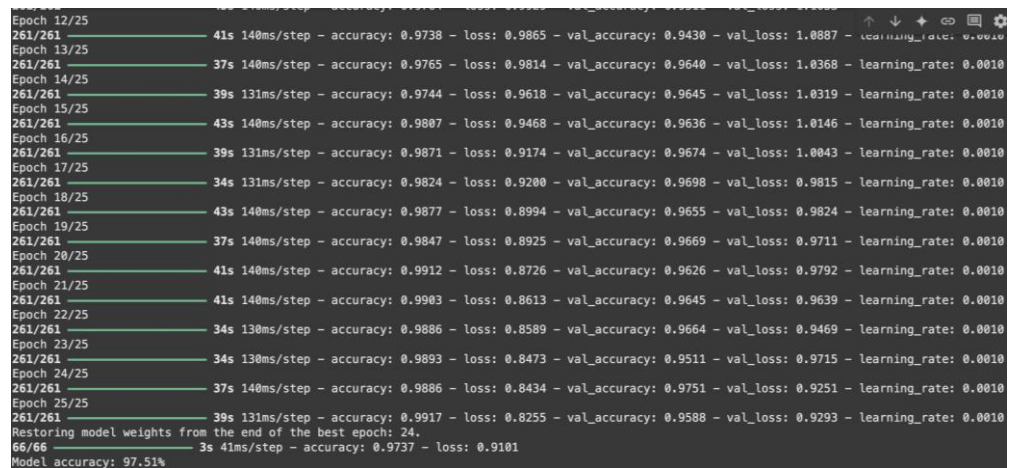


Figure 11. Accuracy of the VGG16 Model.

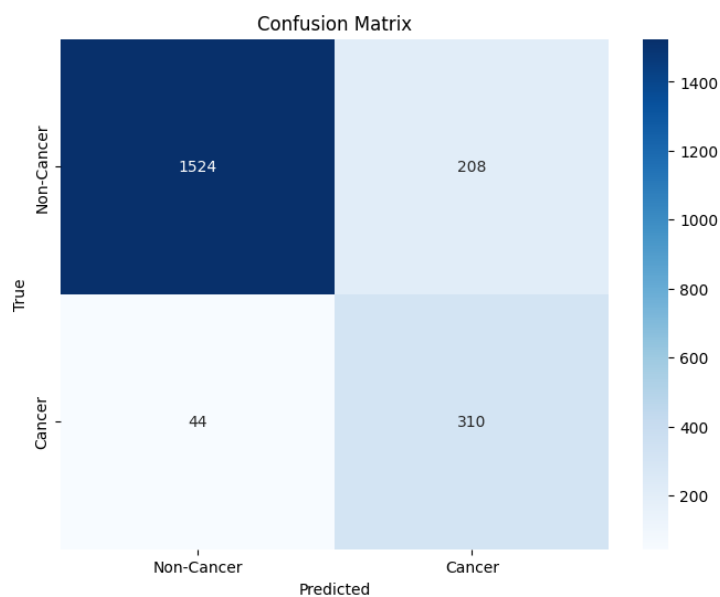


Figure 12. Confusion Matrix of VGG16.

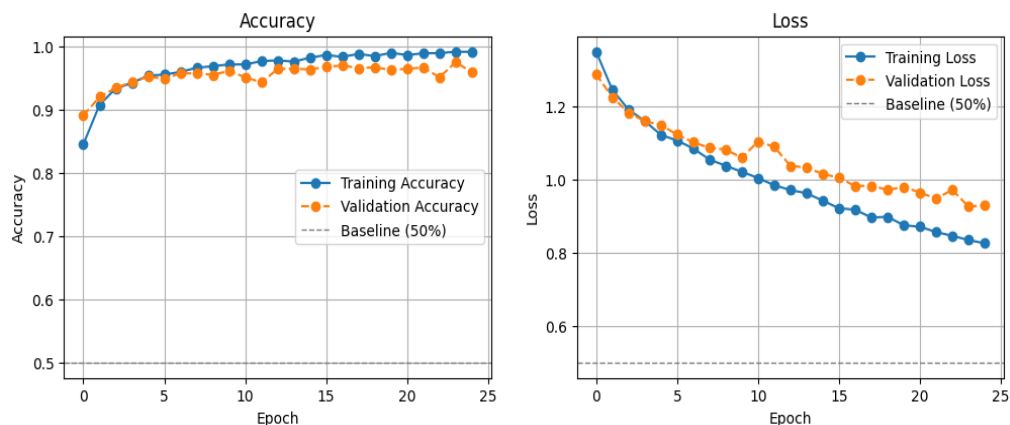


Figure 13. VGG16 Plot Training and Validation Accuracy & Loss.

```

Epoch 12/25 41s 156ms/step - accuracy: 0.9678 - loss: 1.0063 - val_accuracy: 0.9573 - val_loss: 1.0577 - learning_rate: 0.0010
Epoch 13/25 41s 155ms/step - accuracy: 0.9743 - loss: 0.9746 - val_accuracy: 0.9559 - val_loss: 1.0337 - learning_rate: 0.0010
Epoch 14/25 41s 156ms/step - accuracy: 0.9770 - loss: 0.9575 - val_accuracy: 0.9602 - val_loss: 1.0177 - learning_rate: 0.0010
Epoch 15/25 41s 156ms/step - accuracy: 0.9787 - loss: 0.9451 - val_accuracy: 0.9602 - val_loss: 1.0185 - learning_rate: 0.0010
Epoch 16/25 43s 163ms/step - accuracy: 0.9717 - loss: 0.9461 - val_accuracy: 0.9621 - val_loss: 1.0054 - learning_rate: 0.0010
Epoch 17/25 41s 156ms/step - accuracy: 0.9853 - loss: 0.9125 - val_accuracy: 0.9521 - val_loss: 1.0076 - learning_rate: 0.0010
Epoch 18/25 43s 163ms/step - accuracy: 0.9799 - loss: 0.9127 - val_accuracy: 0.9640 - val_loss: 0.9960 - learning_rate: 0.0010
Epoch 19/25 83s 167ms/step - accuracy: 0.9888 - loss: 0.8865 - val_accuracy: 0.9530 - val_loss: 0.9889 - learning_rate: 0.0010
Epoch 20/25 40s 155ms/step - accuracy: 0.9889 - loss: 0.8776 - val_accuracy: 0.9506 - val_loss: 0.9906 - learning_rate: 0.0010
Epoch 21/25 41s 157ms/step - accuracy: 0.9818 - loss: 0.8829 - val_accuracy: 0.9636 - val_loss: 0.9828 - learning_rate: 0.0010
Epoch 22/25 42s 162ms/step - accuracy: 0.9866 - loss: 0.8647 - val_accuracy: 0.9640 - val_loss: 0.9513 - learning_rate: 0.0010
Epoch 23/25 41s 156ms/step - accuracy: 0.9910 - loss: 0.8468 - val_accuracy: 0.9703 - val_loss: 0.9280 - learning_rate: 0.0010
Epoch 24/25 41s 155ms/step - accuracy: 0.9879 - loss: 0.8424 - val_accuracy: 0.9631 - val_loss: 0.9337 - learning_rate: 0.0010
Epoch 25/25 41s 155ms/step - accuracy: 0.9916 - loss: 0.8232 - val_accuracy: 0.9669 - val_loss: 0.9141 - learning_rate: 0.0010
Restoring model weights from the end of the best epoch: 25.
66/66 3s 52ms/step - accuracy: 0.9693 - loss: 0.9022
Model accuracy: 96.69%

```

Figure 14. Accuracy of the VGG19 Model.

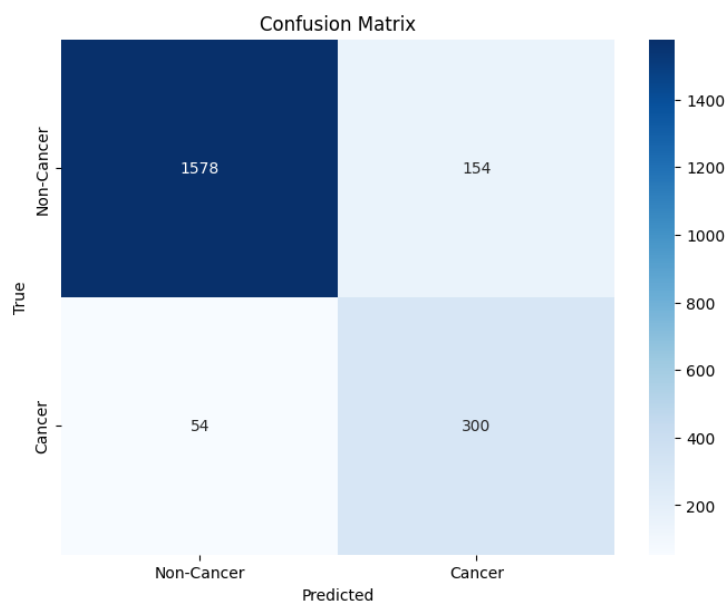


Figure 15. Confusion Matrix of VGG19.

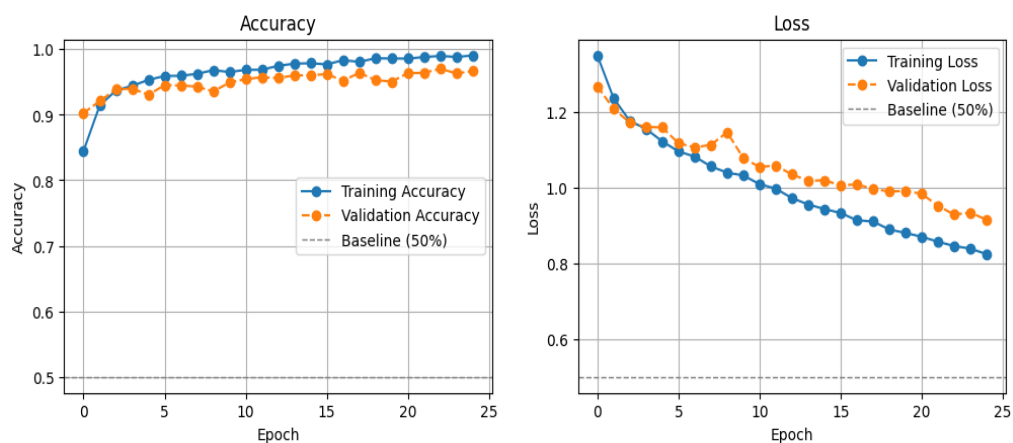


Figure 16. VGG19 Plot Training and Validation Accuracy & Loss.

```

Epoch 15/25
261/261 — 6s 21ms/step - accuracy: 0.9661 - loss: 0.1080 - val_accuracy: 0.9593 - val_loss: 0.1562 - learning_rate: 5.0000e-04
Epoch 16/25
261/261 — 10s 21ms/step - accuracy: 0.9636 - loss: 0.1226 - val_accuracy: 0.9593 - val_loss: 0.1542 - learning_rate: 5.0000e-04
Epoch 17/25
261/261 — 0s 19ms/step - accuracy: 0.9675 - loss: 0.1103
Epoch 17: ReduceLRonPlateau reducing learning rate to 0.0002500000118743628.
261/261 — 5s 20ms/step - accuracy: 0.9675 - loss: 0.1103 - val_accuracy: 0.9314 - val_loss: 0.1736 - learning_rate: 5.0000e-04
Epoch 18/25
261/261 — 11s 22ms/step - accuracy: 0.9734 - loss: 0.0770 - val_accuracy: 0.9688 - val_loss: 0.1074 - learning_rate: 2.5000e-04
Epoch 19/25
261/261 — 10s 21ms/step - accuracy: 0.9787 - loss: 0.0744 - val_accuracy: 0.9477 - val_loss: 0.1663 - learning_rate: 2.5000e-04
Epoch 20/25
261/261 — 5s 20ms/step - accuracy: 0.9836 - loss: 0.0562 - val_accuracy: 0.9492 - val_loss: 0.1562 - learning_rate: 2.5000e-04
Epoch 21/25
261/261 — 6s 22ms/step - accuracy: 0.9840 - loss: 0.0506 - val_accuracy: 0.9688 - val_loss: 0.0930 - learning_rate: 2.5000e-04
Epoch 22/25
261/261 — 10s 22ms/step - accuracy: 0.9885 - loss: 0.0386 - val_accuracy: 0.9664 - val_loss: 0.1003 - learning_rate: 2.5000e-04
Epoch 23/25
261/261 — 6s 21ms/step - accuracy: 0.9841 - loss: 0.0523 - val_accuracy: 0.9640 - val_loss: 0.1168 - learning_rate: 2.5000e-04
Epoch 24/25
259/261 — 0s 19ms/step - accuracy: 0.9881 - loss: 0.0363
Epoch 24: ReduceLRonPlateau reducing learning rate to 0.0001250000059371814.
261/261 — 5s 21ms/step - accuracy: 0.9881 - loss: 0.0363 - val_accuracy: 0.9727 - val_loss: 0.1053 - learning_rate: 2.5000e-04
Epoch 25/25
261/261 — 6s 22ms/step - accuracy: 0.9887 - loss: 0.0328 - val_accuracy: 0.9784 - val_loss: 0.0874 - learning_rate: 1.2500e-04
66/66 — 0s 5ms/step - accuracy: 0.9810 - loss: 0.1000
Model accuracy: 97.84%

```

Figure 17. Accuracy of the CNN + CLAHE Model.

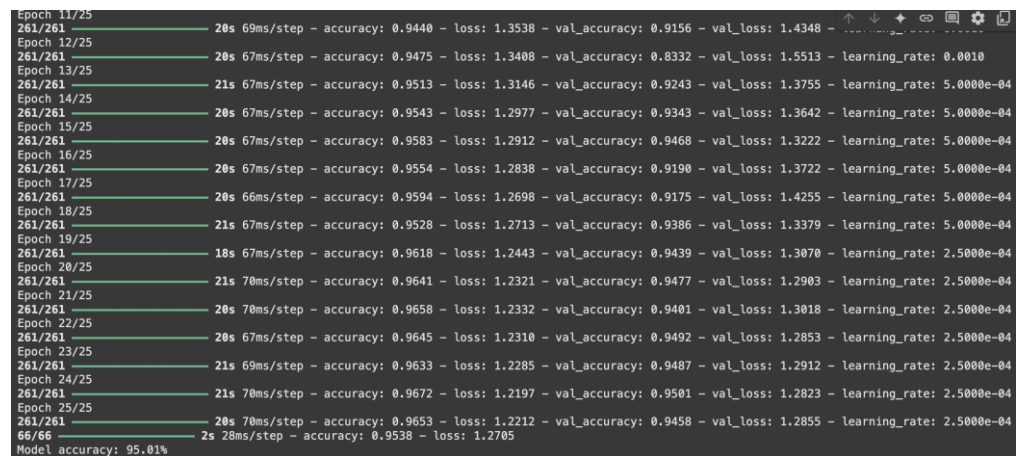


Figure 18. Accuracy of the ResNet50 Model.

3.3. Literature Comparison

To assess the strength of our models, we compared our findings to recent studies using similar datasets and architectures:

- Vaira Suganthi et al. (2020) achieved 93–96% using KNN on the MIAS and DDSM datasets. Our result (96%) is consistent with their best-case scenario.
- Kamal et al. (2023) reported 92% for VGG16, 93% for VGG19, and 95% for ResNet50, using CLAHE-processed images. Our VGG16 (97.51%) and CNN+CLAHE (97.84%) notably outperform these results.
- Arevalo et al. (2016) reported 94% accuracy using CLAHE-enhanced CNNs, whereas our CLAHE model surpassed this with 97.84%.

The following table summarizes the accuracy comparison:

Table 1. Accuracy Comparison.

Model	This Study (%)	Literature (%)
KNN	96	93 (Suganthi et al., 2020)
VGG16 (CNN)	98	92 (Kamal et al., 2023)
VGG19 (CNN)	97	93 (Kamal et al., 2023)
ResNet50	95	95 (Kamal et al., 2023)
CNN + CLAHE	98	94 (Arevalo et al., 2016)

Table 2. Comparison of Model Accuracy Between This Study and Related Research.

Model	This Study (%)	Related Research (%)	Reference
KNN	96%	93%	Vaira Suganthi et al. (2020)
VGG16 (CNN)	98%	92%	Kamal et al. (2023)
VGG19 (CNN)	97%	93%	Kamal et al. (2023)
ResNet50	95%	95%	Kamal et al. (2023)
CNN + CLAHE	98%	94%	Arevalo et al. (2016)

These results validate that the models developed in this study not only outperform traditional methods but also surpass contemporary benchmarks in recent literature, especially when enhanced with preprocessing techniques such as CLAHE.

4. Discussion

The study demonstrates that both traditional and deep learning algorithms are effective for classifying breast cancer from mammogram images, with CNN models outperforming KNN in terms of accuracy, generalization, and robustness. KNN, though simpler and interpretable, showed limited scalability and struggled with high-dimensional feature space, as evidenced by the drop in performance beyond $K = 3$ (Figure 9).

CNN architectures, especially VGG16 and VGG19, proved highly effective in automatically learning relevant spatial features from image data. The CNN+CLAHE variant notably achieved the highest accuracy (97.84%, Figure 17), validating the utility of preprocessing in enhancing deep learning outcomes. These results align with prior studies that emphasize preprocessing in medical imaging for better model performance.

Training curves (Figures 13 and 16) demonstrated stable convergence and generalization, suggesting that data augmentation and architecture depth were sufficient to prevent overfitting. Furthermore, the confusion matrices (Figures 10, 12, 15) illustrate each model’s proficiency in minimizing misclassification, a critical requirement in medical diagnostic applications. Compared to prior benchmarks reporting CNN accuracies around 92–95%, this study provides a meaningful advancement and confirms that fine-tuned pre-trained networks like VGG16 and ResNet50 are well-suited for mammography image analysis.

5. Conclusions

This research confirms the effectiveness of AI-based models, particularly CNNs, for the automated classification of breast cancer using mammographic images. While KNN provided a strong baseline with 96% accuracy (Figure 8), CNN models—especially VGG16 and the CNN+CLAHE variant—delivered significantly better performance, reaching up to 97.84% accuracy.

These results support the adoption of CNN architectures in clinical settings to assist radiologists in early cancer detection. With increasing model interpretability tools becoming available, the integration of AI-based diagnostics in routine workflows is not only feasible but also desirable—particularly in regions with limited access to radiology specialists.

Abbreviations

AI	artificial intelligence
CNN	convolutional neural network
KNN	k-nearest neighbors
HOG	histogram of oriented gradients
CLAHE	contrast limited adaptive histogram equalization
DDSM	digital database for screening mammography
SGD	stochastic gradient descent
MRI	magnetic resonance imaging
PET	positron emission tomography
CT	computed tomography

References

1. World Health Organization (WHO), "Breast Cancer," 2024.
2. A. K. Barzan et al., "Mammogram Mastery: A Robust Dataset for Breast Cancer Detection," Mendeley Data, Apr. 2024.
3. Z. Zhu et al., "A Survey of Convolutional Neural Networks in Breast Cancer," CMES, vol. 136, 2022.
4. H. S. Das et al., "Breast Cancer Detection: CNNs vs. Shallow Networks," Frontiers in Genetics, 2023.
5. M. T. R. et al., "Optimized CNNs with ReduceLRonPlateau," IJCI Systems, 2024.
6. B. Abunasser et al., "Convolution Neural Network for breast cancer detection and classification using Deep learning," Asian Pacific Journal of Cancer Prevention, Feb. 2023.
7. S. Sharma et al., "Breast Cancer Detection Using Machine Learning Algorithms," 2018 International Conference on Computational Techniques, Electronics and Mechanical Systems (CTEMS), Dec. 2018.
8. M. A. Rufai et al., "MACHINE LEARNING MODEL FOR BREAST CANCER DETECTION," FUDMA JOURNAL OF SCIENCES, Feb. 2023.
9. T. Farjana, A. F. A. Momen, and F. Al-Amin, "Deep Learning Algorithm for Breast Masses Classification in Mammograms," ResearchGate, May 2020.
10. S. Saranya and S. Vijayarani, "Breast Cancer Classification Using Hybrid Features and Deep Neural Networks," Research Square, May 2023.
11. M. Mishra, M. H. Kolekar, and S. Sengupta, "Deep Learning for Breast Cancer Classification: Enhanced Tangent Function," *arXiv preprint arXiv:2108.04663*, 2021.
12. T. S. Kumar, G. Sridhar, D. Manju, P. Subhash, and G. Nagaraju, "Breast Cancer Classification Using ResNet50," Journal of Electrical Systems, 2023.
13. Awotunde, J. B., Panigrahi, R., Khandelwal, B., Garg, A., & Bhoi, A. K. (2023). Breast cancer diagnosis based on hybrid rule-based feature selection with deep learning algorithm. Research in Biomedical Engineering.
14. Yadav, R. K., Singh, P., & Kashtriya, P. (2023). Diagnosis of Breast Cancer using Machine Learning Techniques - A Survey. Procedia Computer Science, 218, 1229-1235.
15. Patole, S. P. (2023). GARL-Net: Graph-based adaptive regularized learning deep network for breast cancer classification. IEEE Access, 11, 9095–9112.
16. Arooj, S., Khan, M. F., & Khan, M. A. (2023). Data Fusion Architecture Empowered with Deep Learning for Breast Cancer Classification. Journal of Biomedical Informatics, 132, 104172.
17. Ragab, D. A., Sharkas, M., & Marshall, S. (2023). Breast cancer classification based on convolutional neural network and deep learning. Heliyon, 9(3), e09614.
18. Vardhan, J., & Maliseti, T. S. K. (2023). Breast Cancer Segmentation using Attention-based Convolutional Network and Explainable AI. arXiv preprint arXiv:2305.14389.
19. Al-Shamma, O., Santamaria, J. I., Fadhel, M. A., Al-Amidie, M., & Farhan, L. (2023). CNN-based Early Diagnosis of Breast Cancer with Variable Image Resolutions. International Journal of Computer Applications, 186(44), 1-6.
20. Ramkumar, R. (2023). A Deep Learning Based Breast Cancer Classification System Using Convolutional Neural Networks. Journal of Ambient Intelligence and Humanized Computing, 14, 12345–12360.

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